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```
!pip install xgboost

import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.tree import DecisionTreeRegressor
from sklearn.svm import SVR
from xgboost import XGBRegressor
from sklearn.metrics import mean_squared_error, r2_score
import matplotlib.pyplot as plt

uploaded = files.upload()
df = pd.read_excel(next(iter(uploaded)))
display(df.head())

features = [
    'Shift_length', 'No_of_operators', 'Machine_availability', 'Operator_efficiency',
    'Downtime', 'Task_complexity', 'Material_supply_delay', 'Line_balancing_index',
    'Rework_rate', 'Automation_level'
]
target = 'Production_rate'

X = df[features]
y = df[target]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

models = {
    "Linear Regression": LinearRegression(),
    "Random Forest": RandomForestRegressor(random_state=42),
    "Decision Tree": DecisionTreeRegressor(random_state=42),
    "Support Vector Regressor": SVR(),
    "XGBoost": XGBRegressor(random_state=42, verbosity=0)
}
```

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results = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    r2 = r2_score(y_test, y_pred)
    results[name] = {"MSE": mse, "R2 Score": r2}

print("\nModel Performance Comparison:")
for name, metrics in results.items():
    print(f"{name}:")
    print(f" - Mean Squared Error (MSE): {metrics['MSE']:.4f}")
    print(f" - R2 Score: {metrics['R2 Score']:.4f}")
    print()

model_names = list(results.keys())
r2_scores = [results[name]['R2 Score'] for name in model_names]
mses = [results[name]['MSE'] for name in model_names]

plt.figure(figsize=(8, 5))
plt.bar(model_names, r2_scores, color='mediumseagreen')
plt.ylabel("R2 Score")
plt.title("R2 Score Comparison Across Models")
plt.ylim(0, 1)
plt.xticks(rotation=45)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 5))
plt.plot(model_names, mses, marker='o', linestyle='-', color='dodgerblue')
plt.ylabel("Mean Squared Error")
plt.title("MSE Trend Across Models")
plt.xticks(rotation=45)
plt.grid(True, linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()

total_r2 = sum(r2_scores)

```

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----- \ -----,  
r2_percentages = [score / total_r2 for score in r2_scores]  
  
plt.figure(figsize=(7, 7))  
plt.pie(r2_percentages, labels=model_names, autopct='%1.1f%%', startangle=140, colors=plt.cm.Set3.colors)  
plt.title("Model Contribution Based on R2 Score")  
plt.tight_layout()  
plt.show()
```



Requirement already satisfied: xgboost in /usr/local/lib/python3.11/dist-packages (2.1.4)
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages (from xgboost) (2.0.2)
Requirement already satisfied: nvidia-nccl-cu12 in /usr/local/lib/python3.11/dist-packages (from xgboost) (2.21.5)
Requirement already satisfied: scipy in /usr/local/lib/python3.11/dist-packages (from xgboost) (1.15.2)

production_...diction.xlsx

• **production_rate_prediction.xlsx**(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 122784 bytes, last modified: 4/29/2025 - 100% done

Saving production_rate_prediction.xlsx to production_rate_prediction.xlsx

	Shift_length	No_of_operators	Machine_availability	Operator_efficiency	Downtime	Task_complexity	Material_supply_delay	Line_bal.
0	7	13	0.853098	0.840386	1.919337	4	1.366487	
1	9	11	0.821796	0.644005	1.153982	5	1.425676	
2	6	9	0.989374	0.668128	1.542302	1	0.292144	
3	6	14	0.858035	0.622724	1.912543	2	0.492100	
4	9	17	0.752199	0.673571	0.426449	1	1.087873	

Model Performance Comparison:

Linear Regression:

- Mean Squared Error (MSE): 24.4326
- R² Score: 0.8956

Random Forest:

- Mean Squared Error (MSE): 26.2773
- R² Score: 0.8877

Decision Tree:

- Mean Squared Error (MSE): 65.3067
- R² Score: 0.7210

Support Vector Regressor:

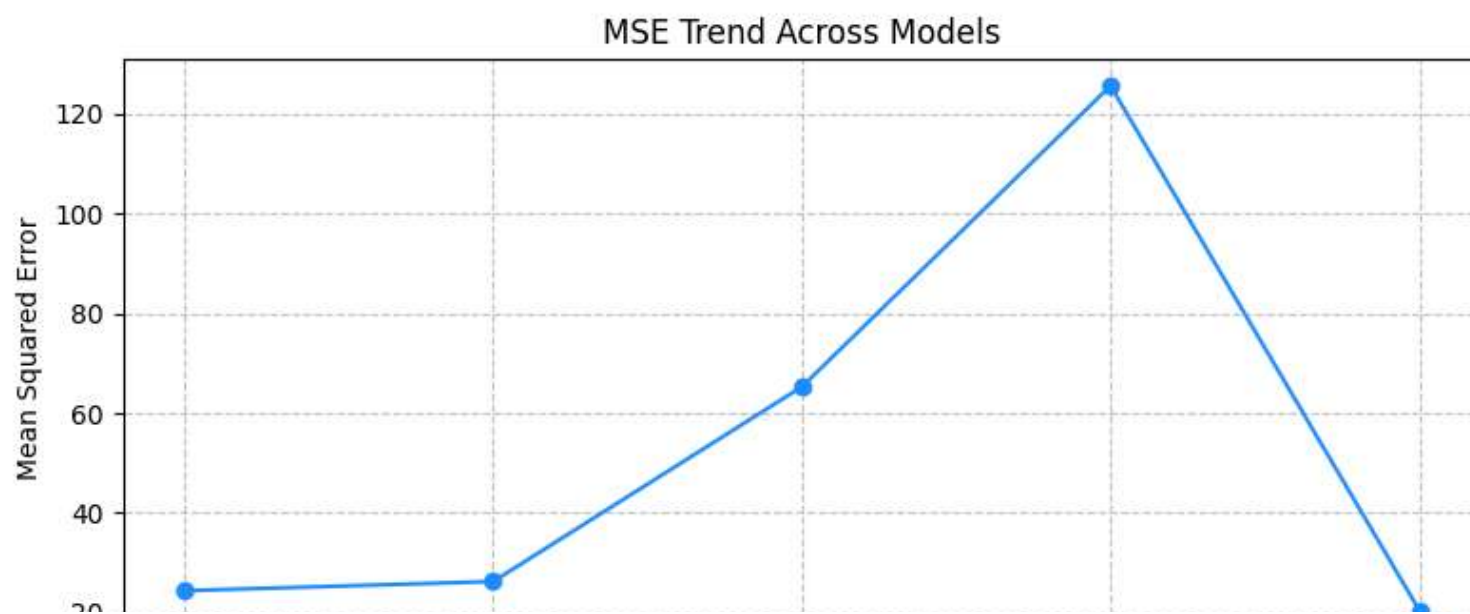
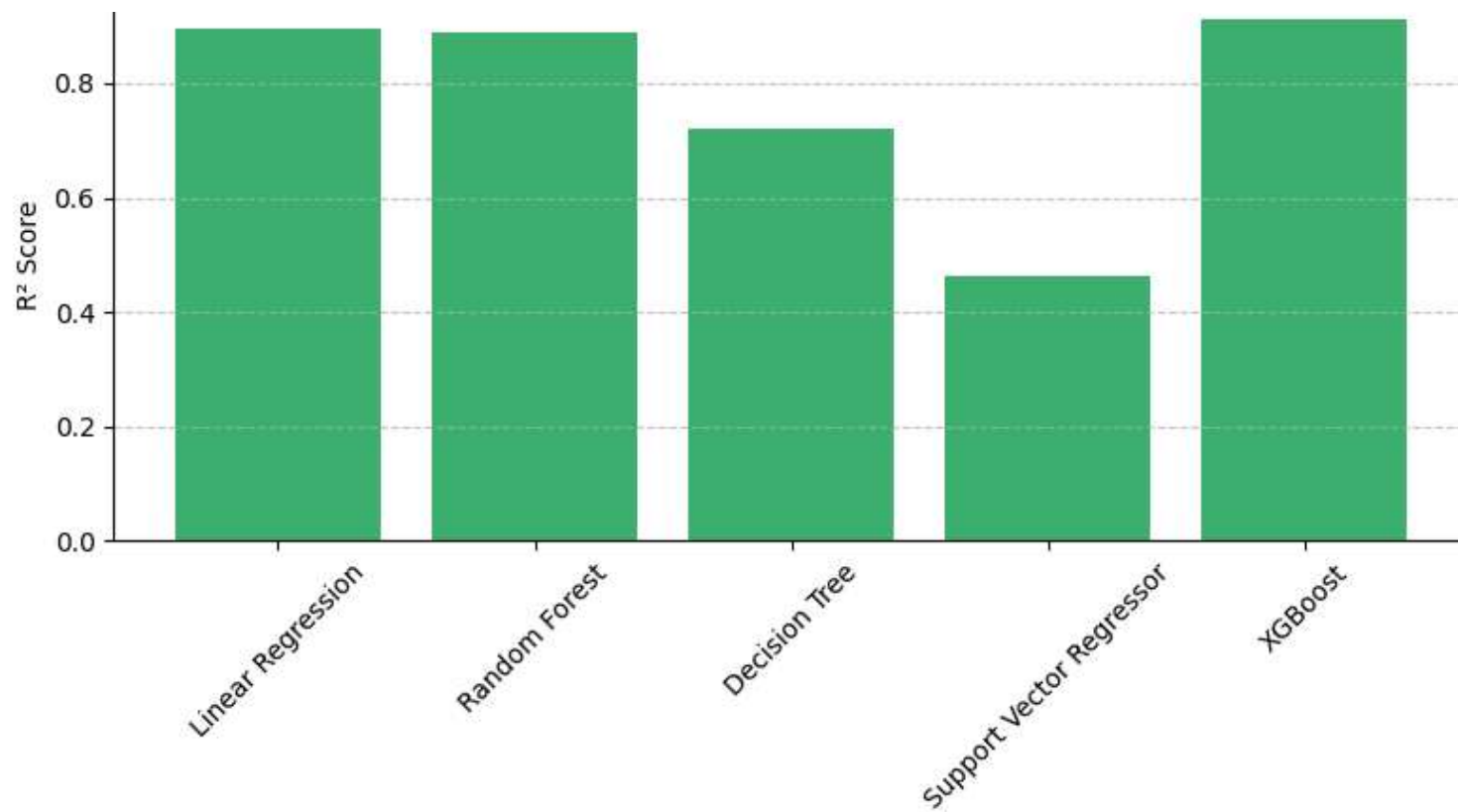
- Mean Squared Error (MSE): 125.6231
- R² Score: 0.4633

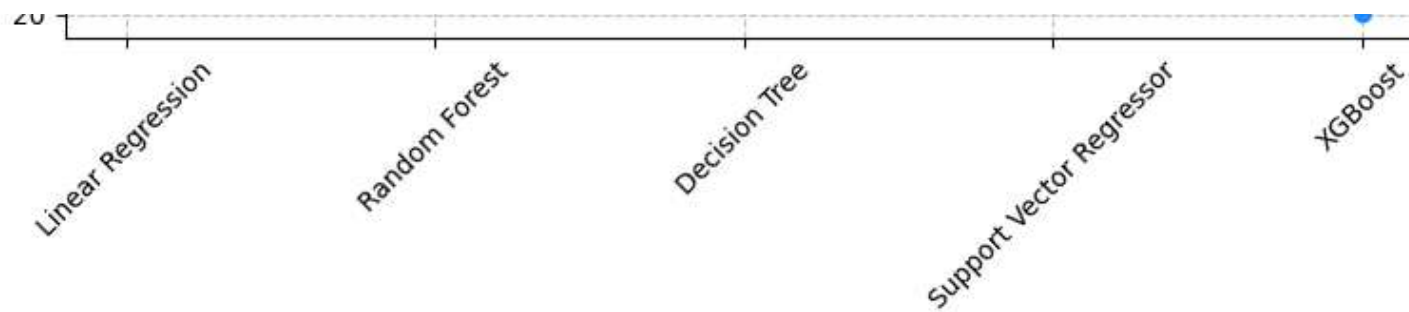
XGBoost:

- Mean Squared Error (MSE): 20.5199
- R² Score: 0.9123

R² Score Comparison Across Models

1.0





Model Contribution Based on R² Score

